Large Scale Social and Complex Networks: Design and Algorithms
ECE 232E Summer 2025
Prof. Vwani Roychowdhury
UCLA, Department of ECE

Syllabus

Course information

- **Instructor:** Professor Vwani Roychowdhury
  - **Email:** vwani@g.ucla.edu
  - **Office hour:** By appointment only
- **Teaching assistant 1:** Kartik Sharma
  - **Email:** kartiksharma@g.ucla.edu
  - **Office hour:** T/Th 6-7 PM over zoom at TBD
  - **Discussion sections:** TBD
- **Teaching assistant 2:** Chenda Duan
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  - **Office hour:** M/W 6-7 PM over zoom at https://ucla.zoom.us/my/chenda
  - **Discussion sections:** TBD

Course prerequisite

- Calculus & Linear Algebra
- Probability
- Basic Python & R skill is required for successful completion of the projects

Course material

- **Lecture notes:** Lecture notes will be uploaded to Bruin Learn on a weekly basis
- **Lecture recordings:** Prerecorded lecture videos can be found under the Echo360 Presentations link on Bruin Learn
- **Discussion material and recordings:** Discussion material will be uploaded to Bruin Learn and the discussion sections will be recorded. The recording will be uploaded to Bruin Learn shortly after the discussion sections end

Lecture schedule

Primary Lecture Set	Description	Projects	Suggested Week to View
EC ENGR 232E-80-Lecture 1	Introduction to Graphs and Complex Networks, definitions of Degree, Path, Cycles; Introduction to Directed Acyclic Graphs (DAG); Motivation for Generative Models using the idea of Information Flow; Power-Law Degree Distribution	1	1
EC ENGR 232E-80-Lecture 2	Paths and Connectedness; Connected Components, Diameter of a Connected Component; Representations of Networks (Adjacency & Incidence Matrix and their space complexity); Multigraphs - Euler Cycles; Trees; Properties of Computations using the different network representations.	1	1
EC ENGR 232E-80-Lecture 3	Properties of Power-Law Networks; Expected values of degree, diameters; Fat-tailed distributions; Generative models for Power-Law networks - Introduction to the Preferential Attachment model	1	2
EC ENGR 232E-80-Lecture 4	Continuation of the discussion on Preferential Attachment model; Random walks and Finding the steady-state of node occupancy in a network by finding the eigenvector/s of the transition probability matrix.	1	2
EC ENGR 232E-80-Lecture 5	Continuation of the discussion on Preferential Attachment model; Doubly preferential models, behavior of the degree distribution as a function of varying the parameters of the generative model.	1	3
EC ENGR 232E-80-Lecture 6	Doubly preferential attachment model; PA model with deletion with Deletion-Compensation protocols; Introduction to Erdos Renyi networks (another generative model) - What is the rate at which the GCC size grows?	1/2	3
EC ENGR 232E-80-Lecture 7	Continuation of the discussion about Erdos Renyi networks; Cayley's theorem; Spanning Trees; What is the likelihood of getting a connected component of size k; Community structure in networks - as a clustering and graph partitioning problem; Modularity; Introduction of the Girvan-Newman algorithm to find communities	1/2	4
EC ENGR 232E-80-Lecture 8	Introduction to Page-Rank; Recap of Eigenvectors as a representation of the steady state node occupancy; teleportation trick; Other variations of page-rank including Personalized Page rank; Discussion of Search Engine Optimization (SEO)	2	4
EC ENGR 232E-80-Lecture 9	Network-driven optimization problems: Random walks; a brief summary of Reinforcement Learning: Markov Chains to Markov Decision Processes, Reward Functions, Value Iterations, Bellman equations, Solving for optimal policy; Value Iteration Algorithm; Inverse Reinforcement Learning Algorithm	2/3	5
EC ENGR 232E-80-Lecture 10	Breadth First Search Algorithms, Dijkstra's lit algorithm; associated proofs	3	5
EC ENGR 232E-80-Lecture 11	Continuing the discussion about the LP feasibility problem - bounding the cost function; Introduction to the max-flow min-cut problem; LP formulation of the problem	3	6
EC ENGR 232E-80-Lecture 12	Continued discussion of the max-flow problem; definition of a cut and using it to compute the max-flow; Planar networks and networks with unit-weighted edges; Matching problem in a bipartite network	3/4	6

Course logistics

- **Assignments:** The course does not contain any written exam. Rather it consists of four equally weighted projects based on real world datasets. The projects can be done in small groups where the group size is capped at 3. You can use Piazza to find group mates

Project	Topic	Post Date	Due Date	Weights
1	Random Graphs and Random Walks	06/23	07/11	25%
2	Social Network Mining	06/23	07/25	25%
3	Reinforcement and Inverse Reinforcement Learning	06/23	08/08	25%
4	Graph Algorithms	06/23	08/29	25%

- **Gradescope Submission:** All the assignment reports need to be submitted via the Gradescope portal in Bruin Learn. Your report must be self-contained, containing all the answers, plots, and necessary code blocks if asked. We will mainly grade the project based on the Gradescope submission. Please make sure you include all your teammates in your Gradescope submission and assign pages correctly, otherwise, it might lead to a 0 for the corresponding project.

- **Bruinlearn Submission:** Please submit all the code (no need for the raw data) necessary to reproduce your project via the Bruinlearn assignment module.

- **Piazza:** Piazza should be the primary means of asking and getting questions answered in the class. We have set up a Piazza platform for the class and you can access and enroll via the Piazza portal in Bruin Learn.